

Long

Q. Explain various data mining techniques?

1. Statistical methods:

⇒ These techniques involve using statistical models and theories to analyze data and make predictions.

- Bayesian inference
- Logistic regression
- ANOVA
- log linear model

2. Cluster analysis:

⇒ Grouping of similar objects based on characteristics.

- Agglomerative algorithm
- Partitional clustering
- incremental clustering
- Divisible algorithm

3. Decision tree & Decision rules:

Used to make decisions based on if/else variable.

- ID3 & C4.5
- CART
- Pruning algorithm

4. Association rules:

⇒ Identifies interesting relationships in large data sets.

- market basket analysis
- Apriori algorithm
- WWW path traversal patterns

5. ANNs:

⇒ Inspired by biological neurons.

- multilayer perceptron (MLP)
- Backpropagation learning
- Kohonen Networks—self organizing maps

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6. GA's:

- ⇒ Inspired by the process of natural selection
- ⇒ solve complex optimization problems.
- ⇒ GA start with a population of potential solutions encoded as chromosomes, and then iteratively evolve & select the best solutions through processes like crossover, mutation and selection.

7. Fuzzy inference systems:

- ⇒ These systems are used to model and reason with uncertainty and imprecision in data.
- ⇒ They are based on fuzzy logic.
- ⇒ Fuzzy inference system use rules and membership functions to make decisions or predictions based on i/p variables

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8. N-dimensional visualization method:

- ⇒ These techniques help visualize high-dimensional data in low lower dimensional space to aid in understanding and analysis.
- Geometric visualization
- Icon-based visualization
- Pixel-oriented visualization
- Hierarchical visualization

